



UNIVERSITY OF BOLOGNA
MASTER'S DEGREE IN COMPUTER SCIENCE AND
ENGINEERING

Lab-activity-03

Laboratory Report

Student:

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0.1 Objective

The objective of this activity is to program a robot in ARGoS using the subsumption architecture. The robot has to find a light source, move towards it, avoid collisions with walls and boxes, and stop when it reaches the black area under the light.

The controller is written in Lua and uses light sensors, proximity sensors, ground sensors and the wheels. The maximum wheel velocity is limited to 15, as required by the assignment.

0.2 Subsumption Architecture

The controller is divided into simple competence levels. Each level receives the wheel speeds produced by the previous level and can either keep them or replace them with its own action. In this way, the controller does not use a central dispatcher, and a new level can be added by inserting another function in the stack.

The implemented levels are:

- **search_light**: base behaviour used when the robot does not perceive the light clearly.
- **go_to_light**: moves the robot towards the side where the light is stronger.
- **avoid_obstacles**: overrides the previous behaviour when an obstacle is detected.
- **halt_on_black**: stops the robot when the target region is reached.

0.3 Behavior Description

At the beginning of each step, the robot starts from the lowest level, **search_light**. This behaviour makes the robot move with a small curve, so that it can explore the arena when the light is not detected.

Then, **go_to_light** reads the light sensors. Sensors from 1 to 12 are used for one side of the robot, while sensors from 13 to 24 are used for the other side. If the total perceived light is too small, the previous search behaviour is kept. Otherwise, the robot turns towards the side where the light value is higher.

The next level is **avoid_obstacles**. It reads the front proximity sensors: sensors from 1 to 4 are used for the left-front side, while sensors from 21 to 24 are used for the right-front side. If the sum of these values is greater than the obstacle threshold, the robot turns away from the more occupied side. This level has higher priority than phototaxis, because avoiding collisions is more important than moving directly towards the light.

The highest level is **halt_on_black**. It reads the ground sensors and stops the robot when at least two sensors detect the black region. Using two sensors makes the stop condition less sensitive to noise.

0.4 Test

The controller was tested with the ARGoS configuration provided for the activity. The tests were executed in headless mode, with a finite duration of 300 seconds, corresponding to 3000 simulation steps.

Two conditions were tested: one without noise, setting actuator and sensor noise to 0.0, and one with noise level 0.02, as suggested in the assignment. For each condition, 10 runs were executed with different random seeds. The measured value was the number of simulation steps needed to reach the black target region.

Noise level	Successful runs	Mean steps	Std. dev.
0.0	5/10	701.4	191.9
0.02	8/10	515.9	184.7

The mean and standard deviation were computed only on successful runs. When the robot did not reach the black region within 3000 steps, the run was counted as unsuccessful.

All simulations completed without Lua or ARGoS runtime errors. The sample is small, so these results should not be interpreted as proof that noise improves the controller. A more reliable evaluation would require more seeds and possibly the final distance from the light for unsuccessful runs.

0.5 Conclusion

The implemented controller follows the main idea of the subsumption architecture. Simple behaviours are placed in a priority stack, where higher levels can suppress the output of lower levels. The robot searches for the light, follows it when it is detected, avoids obstacles, and stops on the black target region.

The solution is intentionally simple and reactive. It does not build a map of the arena and does not use the positioning sensor for control. The positioning sensor can be used only for testing and performance evaluation, as required by the assignment.